

ABHIJIT ROY

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PROFESSIONAL SUMMARY

B.Tech Information Technology graduate (2026) with hands-on, project-based experience building and shipping practical AI applications end-to-end: tool-calling AI agents that reason over natural language to choose between APIs, Retrieval-Augmented Generation (RAG) pipelines, and LLM-powered automation. Comfortable working with open-source and API-based generative AI models (Groq, Mistral AI, Google Gemini), prompt engineering, and agent frameworks (LangChain `create_agent`), with additional hands-on experience in ethical hacking and vulnerability assessment. Active open-source contributor (GitHub) who ships and documents working prototypes independently, and enjoys learning new AI tools and workflows quickly — including generative image/video tools and AI-assisted coding assistants.

SKILLS RELEVANT TO THIS ROLE

Open-Source AI & Generative Models: LLM APIs (Groq, Mistral AI, Google Gemini), prompt engineering, model experimentation, Retrieval-Augmented Generation (RAG), vector databases (ChromaDB), Hugging Face sentence-transformer embeddings

Automation & Architecture: LangChain tool-calling agents (`create_agent`), autonomous multi-tool reasoning, REST API integration (OpenWeatherMap, Tavily), n8n workflow automation, rules-based automation triggers

AI-Assisted Coding & Deployment: Python, Streamlit, FastAPI, Git/GitHub, prototyping and deploying AI applications (Streamlit Community Cloud), reviewing and validating LLM-generated output (Pydantic schema validation)

Cybersecurity: Ethical hacking methodologies, penetration testing, vulnerability assessment and security research (C-DAC internship)

Data Analytics & BI: MS Excel (Advanced), Power BI, SQL, data cleaning and visualization, KPI dashboards

Programming: Python (NumPy, Pandas, Matplotlib), SQL, Java (basic), C (basic), HTML, CSS, JavaScript, React, Node.js

PROJECTS

CityPilot — Tool-Calling AI Agent for City Weather & News

Python · LangChain · Google Gemini · Tavily · OpenWeatherMap API · Streamlit

- Built a multi-tool AI agent that autonomously decides whether to call a live-weather tool, a news-search tool, or both in a single turn, based entirely on reasoning over the user's question with no hardcoded routing logic — directly demonstrating AI agent design and automated multi-tool workflows.
- Integrated external REST APIs (OpenWeatherMap, Tavily) as LangChain tools inside a `create_agent` tool-calling loop, using Google Gemini as the reasoning model.
- Designed a transparent "tool activity" panel in Streamlit that surfaces every tool call, its arguments, and its result, making agent reasoning auditable rather than a black box.

Marginalia — Chat with Your PDF (RAG Chatbot)

Python · LangChain · ChromaDB · Hugging Face Embeddings · Mistral AI · Streamlit

- Built and deployed a full-stack RAG application that lets users upload PDFs and ask natural-language questions, grounding every answer strictly in retrieved document content.
- Engineered the ingestion pipeline — parsing, recursive chunking, and sentence-transformer embeddings — feeding a persistent ChromaDB vector store with MMR-based retrieval.
- Shipped a production-style Streamlit UI with session-isolated document indexing, secure API-key handling, and per-answer source citations.

ReelParse 2026 — AI-Powered Structured Data Extractor

Python · LangChain · Mistral AI (mistral-small-2506) · Pydantic · Streamlit

- Built an AI-powered extractor that parses unstructured text and returns schema-validated structured records using a Pydantic model, replacing fragile string-matching with a production-realistic validation pattern for LLM output.
- Shipped both a Streamlit web UI and a CLI version for interactive and scripted use, and reviewed/validated AI-generated output for correctness before deployment.

CERTIFICATIONS & TRAINING

Data Analytics with Generative AI — Centre of Excellence (COE), Anudip Foundation, Kolkata (Apr 2026 – Jul 2026)

- Trained in combining data analytics techniques with generative AI tools to clean, analyze, and derive insights from datasets.

Ethical Hacking & Penetration Testing Intern — C-DAC, Noida (Aug 2024 – Oct 2024)

- Applied ethical hacking methodologies and penetration testing on real-world systems; documented findings and recommended security improvements.

EDUCATION

B.Tech, Information Technology — Government College of Engineering and Leather Technology, Kolkata | 2022–2026 | CGPA: 7.69/10.0

Class XII (WBCHSE), 2020: 78.4% | Class X (WBBSE), 2018: 80.4%

AWARDS & LANGUAGES

1st Rank, English Debate Competition, Xbridge (2024)

Languages: English (Fluent), Bengali (Fluent), Hindi (Elementary)